Shaheed Zulfikar Ali Bhutto Institute of Science & Technology University

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

Total Marks: 03

Obtained Marks: _____

Mobile App Development

Assignment # 02

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1. Screens and Technical Requirements

Screen	Layout	Purpose
Splash	ConstraintLayout	Logo and app name, then opens Login
Login	LinearLayout	Email/username, password, Login, Register, Forgot Password
Register	LinearLayout	Name, email, password, confirm password with validation
Forgot Password	ConstraintLayout	Accepts email, simulated recovery link
Reset Password	LinearLayout	New and confirm password with validation
Home	GridLayout	6 module cards, each with icon and text
Profile	ConstraintLayout	User information and Logout

Mandatory requirements covered:

- **Resources:** all text in *strings.xml*; day colors in *values/colors.xml* and night-mode colors in *values-night/colors.xml*.
- **Layouts:** LinearLayout, GridLayout and ConstraintLayout are all used.
- **Input controls:** *textEmailAddress*, *textPersonName* and *textPassword*.
- **Validation:** empty fields, email format, minimum password length (6) and password match.
- **Navigation:** explicit Intents between all Activities; Logout clears the back stack.
- **Demo login:** ali@szabist.edu (or username ali) / Ali@1234.

2. Navigation Flow

Splash → Login → Home → Profile → Logout → Login.
Login → Register → Login.
Login → Forgot Password → Reset Password → Login.

3. Screenshots of Main Screen Code

Main (Login) screen: Java Activity and XML layout.

LoginActivity.java

```java


import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class LoginActivity extends AppCompatActivity {

    private EditText etUser, etPassword;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        etUser = findViewById(R.id.etUser);
        etPassword = findViewById(R.id.etPassword);
        Button btnLogin = findViewById(R.id.btnLogin);
        Button btnRegister = findViewById(R.id.btnRegister);
        TextView tvForgot = findViewById(R.id.tvForgot);

        btnLogin.setOnClickListener(v -> attemptLogin());

        btnRegister.setOnClickListener(v ->
                startActivity(new Intent(LoginActivity.this, RegisterActivity.class)));

        tvForgot.setOnClickListener(v ->
                startActivity(new Intent(LoginActivity.this, ForgotPasswordActivity.class)));
    }

    private void attemptLogin() {
        String user = etUser.getText().toString().trim();
        String pass = etPassword.getText().toString();

        etUser.setError(null);
        etPassword.setError(null);

        if (user.isEmpty()) {
            etUser.setError(getString(R.string.error_user_required));
            etUser.requestFocus();
            return;
        }
        if (pass.isEmpty()) {
            etPassword.setError(getString(R.string.error_password_required));
            etPassword.requestFocus();
            return;
        }

        if (UserStore.isValidLogin(user, pass)) {
            Toast.makeText(this, R.string.login_success, Toast.LENGTH_SHORT).show();
            Intent intent = new Intent(LoginActivity.this, HomeActivity.class);
            intent.putExtra("name", UserStore.name);
            startActivity(intent);
            finish();
        } else {
            Toast.makeText(this, R.string.login_failed, Toast.LENGTH_LONG).show();
        }
    }
}
```

activity_login.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<!-- LinearLayout: vertical form -->
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/screen_background"
    android:gravity="center_vertical"
    android:orientation="vertical"
    android:padding="24dp">

    <ImageView
        android:layout_width="72dp"
        android:layout_height="72dp"
        android:layout_gravity="center_horizontal"
        android:contentDescription="@string/app_name"
        android:src="@drawable/ic_logo" />

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:layout_marginBottom="24dp"
        android:gravity="center"
        android:text="@string/login_title"
        android:textColor="@color/text_primary"
        android:textSize="26sp"
        android:textStyle="bold" />

    <EditText
        android:id="@+id/etUser"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:autofillHints="emailAddress"
        android:hint="@string/hint_user"
        android:inputType="textEmailAddress"
        android:minHeight="48dp"
        android:textColor="@color/text_primary"
        android:textColorHint="@color/text_secondary" />

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:autofillHints="password"
        android:hint="@string/hint_password"
        android:inputType="textPassword"
        android:minHeight="48dp"
        android:textColor="@color/text_primary"
        android:textColorHint="@color/text_secondary" />

    <TextView
        android:id="@+id/tvForgot"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="end"
```

```
        android:layout_marginTop="8dp"
        android:padding="8dp"
        android:text="@string/forgot_password"
        android:textColor="@color/brand_primary"
        android:textStyle="bold" />

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="52dp"
        android:layout_marginTop="16dp"
        android:text="@string/btn_login" />

    <Button
        android:id="@+id/btnRegister"
        style="@style/Widget.MaterialComponents.Button.OutlinedButton"
        android:layout_width="match_parent"
        android:layout_height="52dp"
        android:layout_marginTop="12dp"
        android:text="@string/btn_register" />

</LinearLayout>
```

Home screen (modules grid) and resources.

➢ HomeActivity.java

```
package com.example.commonscreens;

import android.content.Intent;
import  android.os.Bundle;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
import androidx.cardview.widget.CardView;

public class HomeActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_home);

        String name = getIntent().getStringExtra("name");
        TextView tvWelcome = findViewById(R.id.tvWelcome);
        tvWelcome.setText(getString(R.string.welcome_user,  name));

        setupModule(R.id.cardCourses, R.string.module_courses);
        setupModule(R.id.cardAssignments, R.string.module_assignments);
        setupModule(R.id.cardAttendance, R.string.module_attendance);
        setupModule(R.id.cardResults, R.string.module_results);
        setupModule(R.id.cardNotices, R.string.module_notices);

        // Profile card opens the Profile screen with an explicit Intent
        CardView cardProfile = findViewById(R.id.cardProfile);
        cardProfile.setOnClickListener(v ->
                startActivity(new  Intent(HomeActivity.this,  ProfileActivity.class)));
    }

    private void setupModule(int cardId, int nameRes) {
        CardView card = findViewById(cardId);
        card.setOnClickListener(v ->
                Toast.makeText(this, getString(R.string.module_opened, getString(nameRes)),
                        Toast.LENGTH_SHORT).show());
    }
}
```

- **activity_home.xml (GridLayout with module cards, first part)**

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/screen_background"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/home_title"
        android:textColor="@color/text_primary"
        android:textSize="28sp"
        android:textStyle="bold" />

    <TextView
        android:id="@+id/tvWelcome"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp"
        android:textColor="@color/text_secondary"
        android:textSize="16sp" />

    <!-- GridLayout: 2 columns x 3 rows of module cards -->
    <GridLayout
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:columnCount="2"
        android:rowCount="3">

        <androidx.cardview.widget.CardView
            android:id="@+id/cardCourses"
            style="@style/ModuleCard">
            <LinearLayout style="@style/ModuleContent">
                <ImageView style="@style/ModuleIcon" android:src="@drawable/ic_courses"
                    android:contentDescription="@string/module_courses" />
                <TextView style="@style/ModuleLabel" android:text="@string/module_courses" />
            </LinearLayout>
        </androidx.cardview.widget.CardView>

        <androidx.cardview.widget.CardView
            android:id="@+id/cardAssignments"
            style="@style/ModuleCard">
            <LinearLayout style="@style/ModuleContent">
                <ImageView style="@style/ModuleIcon" android:src="@drawable/ic_assignments"
                    android:contentDescription="@string/module_assignments" />
                <TextView style="@style/ModuleLabel" android:text="@string/module_assignments" />
            </LinearLayout>
```

Day and night-mode color resources.

- **res/values/colors.xml (default)**

```xml
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <!-- Default (day) colors -->
    <color name="brand_primary">#1565C0</color>
    <color name="brand_primary_dark">#0D47A1</color>
    <color name="brand_accent">#00ACC1</color>
    <color name="screen_background">#F4F7FB</color>
    <color name="card_background">#FFFFFF</color>
    <color name="text_primary">#1B1F2A</color>
    <color name="text_secondary">#5F6B7A</color>
    <color name="on_brand">#FFFFFF</color>
</resources>
```

res/values-night/colors.xml (night mode)

```xml
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <!-- Alternative (night mode) colors -->
    <color name="brand_primary">#64B5F6</color>
    <color name="brand_primary_dark">#1E88E5</color>
    <color name="brand_accent">#4DD0E1</color>
    <color name="screen_background">#10141C</color>
    <color name="card_background">#1C2330</color>
    <color name="text_primary">#EAEFF7</color>
    <color name="text_secondary">#A3AEC0</color>
    <color name="on_brand">#0B1220</color>
</resources>
```

RegisterActivity.java (validation part)

```java
        btnCreate.setOnClickListener(v -> attemptRegister());

        tvBack.setOnClickListener(v -> {
            startActivity(new Intent(RegisterActivity.this, LoginActivity.class));
            finish();
        });
    }

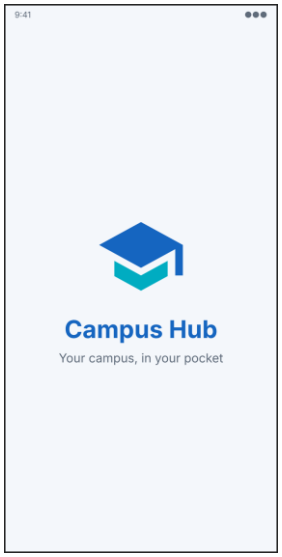
    private void attemptRegister() {
        String name = etName.getText().toString().trim();
        String email = etEmail.getText().toString().trim();
        String pass = etPassword.getText().toString();
        String confirm = etConfirm.getText().toString();

        if (name.length() < 3) {
            etName.setError(getString(R.string.error_name));
            etName.requestFocus();
            return;
        }
        if (!Patterns.EMAIL_ADDRESS.matcher(email).matches()) {
            etEmail.setError(getString(R.string.error_email));
            etEmail.requestFocus();
            return;
        }
        if (pass.length() < 6) {
            etPassword.setError(getString(R.string.error_password_length));
            etPassword.requestFocus();
            return;
        }
        if (!pass.equals(confirm)) {
            etConfirm.setError(getString(R.string.error_password_match));
            etConfirm.requestFocus();
            return;
        }

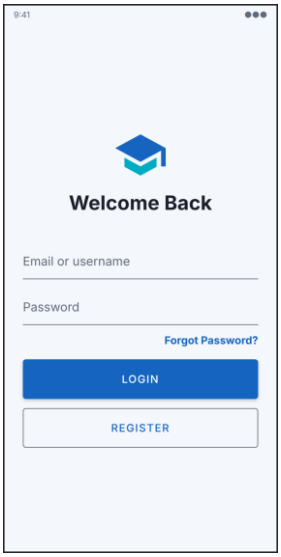
        UserStore.register(name, email, pass);
        Toast.makeText(this, R.string.register_success, Toast.LENGTH_LONG).show();
        startActivity(new Intent(RegisterActivity.this, LoginActivity.class));
```

4. Screenshots of Application Output

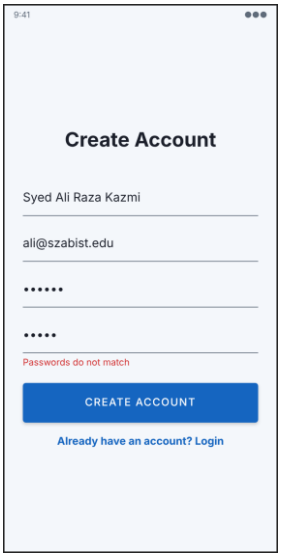
Sample UI previews *of the app screens (rendered from the app's layouts, strings and day/night colors; not emulator captures).*



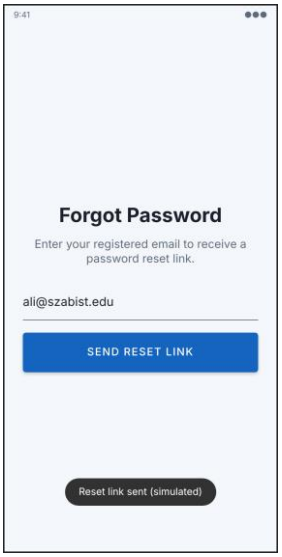
Splash



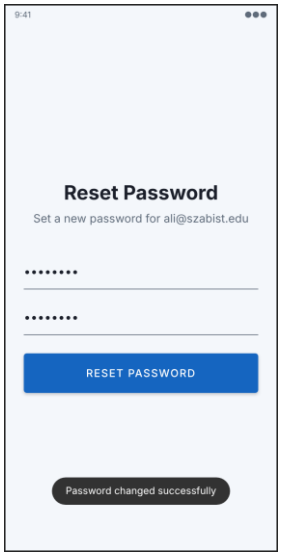
Login



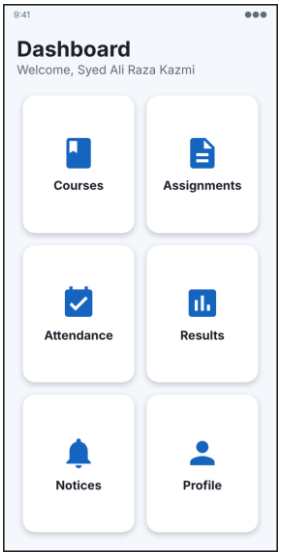
Register



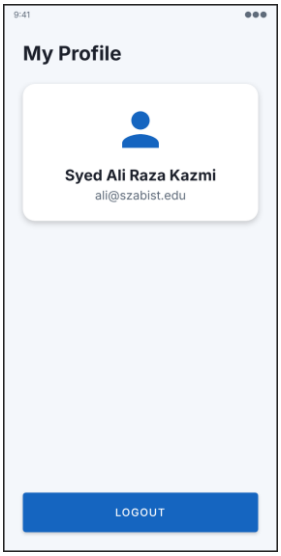
Forgot Password



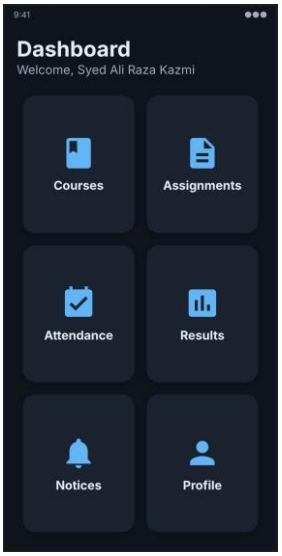
Reset Password



Home



Profile / Logout



Home (night mode)

Sample UI preview – replace with emulator screenshots after running the app in Android Studio.

5. GitHub Repository

https://gist.github.com/salikazmj2-eng